



Figure 1. Differences between planned and measured dose with micro-MOSFET dosimeters in patients.

large differences (> 10%) between planned and measured dose to verify the applicator configuration. Potentially, patients can be identified who may benefit from re-planning to improve overall treatment quality.

#### PO134

##### Evaluation of Dose Response in 106Ru Eye Plaque Brachytherapy Using a Novel Software Tool

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**Purpose:** A software tool for 106Ru brachytherapy of uveal melanoma was inhouse developed that allows three dimensional dose calculations for the tumor and various ophthalmic organs at risk. This software is based on a flexible eye model and pre-calculated dosimetric lookup tables for different types of BEBIG eye plaques (CCA, CCB and COB), generated from MC simulations using MCNP6. The aim of the study was to evaluate dose-response relationships using this software package and to determine prognostic values for tumor control and morbidity when treating uveal melanomas with 106Ru eye brachytherapy.

**Materials and Methods:** 39 patients undergoing 106Ru eye plaque brachytherapy were included in this retrospective study. Individual recalculation of dose distributions was performed using a re-sizable 3D eye model created with the software package Sidefx Houdini. The eye model consists of a dome-shaped tumor model, where apex height and basal diameter are adjustable parameters, and various critical structures (e.g. lens, ciliary body, optic nerve, macula, retina and sclera). The position and shape of the tumor was modelled using the above degrees of freedom and reported data on the distance between tumor and macula as well as tumor and optic nerve. Additionally, the fundus images were projected onto the 3D eye model in order to account for the individual tumor shape. For each patient the dose distribution was based on MC generated dose libraries of BEBIG eye plaques (CCA, CCB and COB), that were individually positioned. Finally, dose and dose volume parameters as well as dose volume histograms (DVH) were generated for the tumor and the various organs at risk. Tumor coverage was reported in terms of D98; the dose to the retina was reported as Dmean and correlation with retinopathy examined. Tumor control and morbidity correlations were checked with a two-sample t-test. For quality assurance purposed film measurements were performed to provide an estimation of the dosimetric uncertainties and safety margins inherent to 106Ru brachytherapy of uveal melanoma.

**Results:** The software tool enabled to calculate full 3D dose distributions on an adjustable 3D eye model with a dose calculation grid of 200 µm. For the 39 patients analyzed in this study the local tumor control was 95%. A

statistically significant correlation between tumor coverage in terms of D98 ( $p = 0.013$ ) was observed. In our patient cohort seven patients developed retinopathy, which was in turn found to be significantly influenced by the mean dose to the retina ( $p = 0.021$ ). Safety margins can be evaluated considering the dosimetric implications.

**Conclusions:** A novel software tool for treatment planning in 106Ru brachytherapy of uveal melanoma was developed, that enables retrospective studies for dose response analysis. Furthermore it is flexible enough to support treatment planning in prospective studies, with treatment planning features common in external beam therapy, such as tumor and organ at risk segmentation, three dimensional dose calculation and dose-volume metrics. Future work will try to extend the retrospective analysis on the basis of a larger patient cohort to provide a broader range of prognostic values.

#### PO135

##### On the Utility of Pre Treatment kV-CBCT for MRI-Based Planning of Cervical Cancer Patients

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**Purpose:** MRI-based treatment planning has become the gold standard for HDR treatment of cervical cancer. In most clinical implementations, there is a span of 2-4 hours between applicator insertion/imaging and treatment. Moreover, MRI machines are rarely part of the HDR treatment delivery room, meaning that patients are moved several times before the actual treatment. The purpose of this study is to expand an initial pilot study meant to determine if it is feasible to use updated (adapted) doses to organs at risk (OAR) as determined by a pre-treatment kV-CBCT imaging performed with patient in treatment position.

**Materials and Methods:** Cervical cancer patients undergoing external beam radiotherapy followed by an HDR boost were included in this study. MRI (1.5T GE Excite, GE Healthcare) and kV-CBCT (Acuity, Varian Medical Corporation) imaging were performed after applicator insertion. After image registration, applicator delineation (on kV-CBCT, 2.5 mm acquisition, 1 mm reconstruction), contouring of HRCTV and IRCTV and normal tissue (on MRI), a treatment plan was generated that met the constraints imposed by the GEC-ESTRO guidelines. Before treatment, another kV-CBCT (2.5 mm slice thickness) was performed with patient in treatment position (preTX). Normal tissues were recontoured on kV-CBCT. The treatment plan was also reproduced on the preTX CBCT and the dose to 2 cm<sup>3</sup> for bladder, rectum, sigmoid, bowel, and vagina were compared between planning and preTX using a two-sided paired t test. P values ≤0.05 were considered significant. In addition, for each OAR, we calculated the percentage of instances with difference between planning and preTX >10%, 20% respectively, and the percent of cases where these differences were positive, meaning the dose calculated on the preTX scan is larger than initially estimated by the treatment plan. This scenario was considered as it depicts the cases where reported dose would be potentially underestimated.

**Results:** Twenty cervical cancer patients were included for a total of 78 fractions. Table 1 shows the summary statistics for all OARs for the treatment plan (Plan) and the CBCT-based preTX plan. The image quality generated with kV-CBCT is known to suffer from reduced imaging contrast, increased cupping, streaking artifacts, and less accurate HUs. While the last shortcoming is not important in dose calculations based on TG43 formalisms, the reduction in image quality leads to uncertainty in bowel/uterus and bladder/uterus interface, superior-inferior delineation between rectum and sigmoid, and bowel/sigmoid identification, leading to large variations in contours when compared to MRI. Statistically significant differences were found for bladder and rectum for which in almost half the cases the differences in the dose to 2 cm<sup>3</sup> of organ were greater than 10%, and in approximately 40% of cases the differences in dose were greater than 20%. The main two reasons identified for the large variations are: 1) differences between kV-CBCT and MR in soft tissue